

SERIES AND SEQUENCES

Objectives:

- After reading this book, you should be able to
- Recognize the difference between a sequence and series
 - Recognize an arithmetic progression
 - Find the n^{th} term of an arithmetic progression
 - Find the sum of an arithmetic progression
 - Recognize a geometric progression
 - Find the n^{th} term of a geometric progression A. P
 - Find the sum to infinity of an A.P with common ratio when $|r| < 1$
 - Understand Σ notation for sum of series
 - Be familiar with standard formula for $\Sigma r, \Sigma r^2, \Sigma r^3$.
 - Prove expressions by mathematical induction.
 - Apply geometric and arithmetic progression to solve word problems.

Introduction

A **sequence** is a set of number stated in a definite order such that each number forming a set can be obtained from the previous one according to some rule.

For example

- (a) $\{2, 4, 6, 8, 10 \dots\}$
 (b) $\{1, -1, 1, -1, 1 \dots\}$
 (c) $\{10, 11, 9, 12, 8, 13 \dots\}$
 (d) $\{1, 4, 9, 16, 25 \dots\}$
 (e) $\left\{8, -4, 2, -1, \frac{1}{2} \dots\right\}$
 (f) $\{1, 1, 2, 3, 5, 8 \dots\}$

A **series** is obtained by adding up all the terms of the sequence for example. Suppose we have a sequence

$$u_1, u_2, u_3, \dots, u_n$$

The series can be obtained from the above sequence. The series is $u_1 + u_2 + u_3 + \dots + u_n$

From the examples of sequences above. It is possible to give a formula for the general or n^{th} term of each of the above series.

Thus for a, $u_n = 2n$, where $n = 1, 2, 3, \dots$

For b, $u_n = (-1)^{n+1}$ where $n = 1, 2, 3, \dots$

For c, u_n is not easy to find but we can give a separate formulae for odd and even terms of the sequence.

$$u_{2n} = 10 + n$$

$$u_{2n+1} = 10 - n$$

For d $u_n = n^2$ where $n = 1, 2, 3, \dots$

For e, the formula u_n is not obvious

To obtain each successive term we divide by 2 then change sign. Thus at each stage we multiply by $\left(-\frac{1}{2}\right)$ which means that

$$u_n = 8 \left(-\frac{1}{2}\right)^{n-1}$$

For f, each term is a sum of two previous terms. This can be expressed formally as a relation between u_n and u_{n-1}, u_{n-2} ($u_n = u_{n-1} + u_{n-2}$). In this case more advanced technique are needed to find the formula.

If the sequence ends after a certain number of terms it's said to be **finite**. A sequence which continues indefinitely is said to be **infinite**.

ARITHMETIC PROGRESSION (A.P)

Arithmetic progression is a series in which one term is obtained from the previous one by adding a constant number.

For example:

- (a) $1 + 2 + 3 + 4 + \dots + 98 + 99$
 (b) $6 + 10 + 14 + 18 + \dots + 46 + 50$
 (c) $10 + 7 + 4 + 1 + \dots - 47 - 50$

The constant number is called a common difference. In the above examples the common differences are 1 for a, 4 for b and -3 for c respectively.

The arithmetic progression is completely defined when the first term (**a**) and common difference, (**d**) are given

Generally, the arithmetic progression series is given by

$$a + (a + d) + (a + 2d) + \dots + a + (n - 1)d$$

$a = \text{first term}$

$d = \text{common difference}$

$a + (n - 1)d = n^{\text{th}} \text{ term}$

Example 1

Which of the following series are arithmetic progression.

Write down the common difference of those that are

(a) $7 + 11 + 15 + \dots$

(b) $-7 - 5 - 3 + \dots$

(c) $-17 - 12 - 7 + \dots + 13$

(d) $1 + 1\frac{1}{4} + 1\frac{1}{2} + \dots 9\frac{3}{4}$

(e) $1^2 + 2^2 + 3^2 + 4^2 \dots$

Solution

(a) $7 + 11 + 15 + \dots$

$$a = 7$$

$$d_1 = 11 - 7 = 4$$

$$d_2 = 15 - 11 = 4$$

$$\Rightarrow d_1 = d_2 = d = 4 (\text{common difference})$$

$$\Rightarrow 7 + 11 + 15 + \dots$$

Is an arithmetic progression with first term $a = 7$ and common difference $d = 4$

(b) $-7 - 5 - 3 + \dots$

$$a = -7, \quad d_1 = -5 - -7 = 2$$

$$d_2 = -3 - -5 = 2$$

$$d_1 = d_2 = d = 2 \quad (\text{common difference})$$

It is an arithmetic progression with first term $a = -7$ and common difference 2

(c) $-17 - 12 - 7 + \dots 13$

$$a = -17$$

$$d_1 = -12 - -17 = 5$$

$$d_2 = -7 - -12 = 5$$

It is an arithmetic progression with the first term $a = -17$ and common difference (d) = 5

(d) $1 + \frac{5}{4} + \frac{3}{2} + \dots \frac{39}{4}$

$$a = 1$$

$$d_1 = \frac{5}{4} - 1 = \frac{1}{4}$$

$$d_2 = \frac{3}{2} - \frac{5}{4} = \frac{1}{4}$$

$$a = 1, \quad d_1 = d_2 = d = \frac{1}{4}$$

$$\Rightarrow 1 + \frac{5}{4} + \frac{3}{2} + \dots \frac{39}{4} \text{ is an arithmetic progression}$$

(e) $1^2 + 2^2 + 3^2 + 4^2 \dots$

$$a = 1^2 = 1$$

$$d_1 = 2^2 - 1^2 = 3$$

$$d_2 = 3^2 - 2^2 = 5$$

$$d_3 = 4^2 - 3^2 = 7$$

$$d_1 \neq d_2 \neq d_3.$$

$1^2 + 2^2 + 3^2 + 4^2 + \dots$ is not an arithmetic progression.

Example II

Write down the terms indicated in each of the following arithmetic progression

(i) $2 + 6 + 10 + 14 + \dots 12^{\text{th}}$

(ii) $10 + 8 + 6 + 4 \dots 15^{\text{th}}$

(iii) $10 + 8 + 6 + 4 + \dots$

(iv) $-6\frac{1}{2} - 5 - 3\frac{1}{2} - 2 + \dots 12^{\text{th}}$

Solution

$$2 + 6 + 10 + 14 + \dots 12^{\text{th}}$$

The n^{th} term of an A.P (u_n) is given by:

$$u_n = a + (n - 1)d$$

$$a = 2, \quad d = 4$$

$$\Rightarrow u_n = 2 + (n - 1)4$$

$$u_n = 2 + 4n - 4$$

$$u_n = 4n - 2$$

$$12^{\text{th}} \text{ term, } n=12$$

$$\Rightarrow u_{12} = 4 \times 12 - 2$$

$$= 48 - 2$$

$$= 46$$

(ii) $10 + 8 + 6 + 4 \dots 15^{\text{th}}$

$$n^{\text{th}} \text{ term, } u_n = a + (n - 1)d$$

$$u_{15} = 10 + 14 \times -2$$

$$= 10 - 28$$

$$= -18$$

(ii) $7 + 3 - 1 - 5 + \dots 19^{\text{th}}$

$$a = 7$$

$$d = 3 - 7 = -4$$

$$u_n = a + (n - 1)d$$

$$u_{19} = 7 + (19 - 1)(-4)$$

$$= 7 + -72$$

$$u_{19} = -65$$

(iv) $-6\frac{1}{2} + -5 - 3\frac{1}{2} - 2 + \dots$

$$a = -6.5$$

$$d = 1.5$$

$$n^{\text{th}} \text{ term } u_n = a + (n - 1)d$$

$$u_n = -6.5 + (n - 1)(1.5)$$

$$u_{12} = -6.5 + (12 - 1) \times 1.5$$

$$= -6.5 + 16.5$$

$$= 10$$

Example III

Find the number of terms in each of the following progression

- (a) $5 + 8 + 11 + 14 + \dots 59 + 62$
 (b) $1 + 6 + 11 + 16 + \dots 501 + 506$
 (c) $-193 - 189 - 185 + \dots - 21 - 17$
 (d) $2\frac{1}{4} + 2\frac{17}{20} + 3\frac{9}{20} + \dots 20\frac{1}{4} + 20\frac{17}{20}$

Solution

$$5 + 8 + 11 + 14 + \dots 59 + 62$$

$$a = 5 \quad d = 8 - 5 = 3$$

$$n^{\text{th}} \text{ term} = a + (n - 1)d$$

$$62 = a + (n - 1)d$$

$$62 = 5 + (n - 1)3$$

$$62 - 5 = 3(n - 1)$$

$$\frac{57}{3} = n - 1$$

$$20 = n$$

(b) $1 + 6 + 11 + 16 + \dots 501 + 506$

$$a = 1 \quad d = 6 - 1 = 5$$

$$n^{\text{th}} \text{ term } u_n = a + (n - 1)d$$

$$u_n = 1 + (n - 1)5$$

$$506 = 1 + (n - 1)5$$

$$505 = 5(n - 1)$$

$$101 = n - 1$$

$$n = 102$$

(c) $-193 + -189 + -185 + \dots - 21 + -17$

$$a = -193$$

$$d = -189 - -193$$

$$d = 4$$

The n^{th} term

$$u_n = a + (n - 1)d$$

$$u_n = -193 + (n - 1)4$$

$$-17 = -193 + (n - 1)4$$

$$-17 + 193 = 4(n - 1)$$

$$\frac{176}{4} = n - 1$$

$$n = 45$$

Example IV

An AP is given by $k, \frac{2k}{3}, \frac{k}{3}, 0 \dots$

(i) Find the sixth term

(ii) Find the n^{th} term

(iii) If the 20^{th} term is 15. Find K

Solution

$$a = k, d = \frac{2k}{3} - k = \frac{-k}{3}$$

$$n^{\text{th}} \text{ term } (u_n) = k + (n - 1)\left(\frac{-k}{3}\right)$$

$$6^{\text{th}} \text{ term } (u_6) = k + 5\left(\frac{-k}{3}\right)$$

$$= k - \frac{5k}{3}$$

$$= -\frac{2k}{3}$$

$$(ii) n^{\text{th}} \text{ term} = k + (n - 1)\left(-\left(\frac{k}{3}\right)\right)$$

$$= k - \frac{nk}{3} + \frac{k}{3}$$

$$= \frac{4k}{3} - \frac{nk}{3}$$

$$= \frac{k}{3}(4 - n)$$

But $u_{20} = 15$

$$\Rightarrow 15 = \frac{k}{3}(4 - 20)$$

$$45 = k(-16)$$

$$k = \frac{-45}{16}$$

Sum of Arithmetic Progression

Generally, the arithmetic progression is given by

$$a + (a + d) + (a + 2d) + \dots a + (n - 2)d + a + (n - 1)d$$

Let the sum of the arithmetic progression be S_n

$$S_n = a + (a + d) + (a + 2d) + \dots a + (n - 2)d + a + (n - 1)d \dots \dots \dots (1)$$

Suppose the terms are added in opposite order

$$S_n = a + (n - 1)d + a + (n - 2)d + \dots (a + 2d) + (a + d) + a \dots \dots \dots (2)$$

adding eqn (1) and (2)

$$\Rightarrow 2S_n = 2a + (n - 1)d + 2a + (n - 1)d$$

$$+ 2a + (n - 1)d + \dots 2a + (n - 1)d$$

$$+ 2a + (n - 1)d + 2a + (n - 1)d \dots \dots \dots (3)$$

Since there are n terms of equation (3) that have been added together. The total is $2S_n = n(2a + (n - 1)d)$

$$\Rightarrow 2S_n = n(2a + (n - 1)d)$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$\boxed{S_n = \frac{n}{2}(2a + (n - 1)d)}$$

This is the formula for the sum of arithmetic progression

Example I

Find the sum of the first 50 terms of an A.P

$$1 + 3 + 5 + 7 + 9 + \dots$$

Solution

$$a = 1, d = 2$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{50} = \frac{50}{2}(2 \times 1 + 49 \times 2)$$

$$S_{50} = 25(2 + 98)$$

$$S_{50} = 25(100)$$

$$S_{50} = 2500$$

Example II

Find the sum of $5 + 9 + 13 + \dots$ to 20 terms

Solution

$$a = 5, d = 4$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{20} = \frac{20}{2}(2 \times 5 + (20 - 1) \times 4)$$

$$S_{20} = 10(10 + 19 \times 4)$$

$$S_{20} = 10(86)$$

$$S_{20} = 860$$

Example III

Find the sum $27 + 22 + 17 + \dots$ to 10 terms

Solution

$$a = 27, d = 22 - 27$$

$$d = -5$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{10} = \frac{10}{2}(2 \times 27 + (10 - 1) \times -5)$$

$$S_{10} = \frac{10}{2}(54 + -45)$$

$$S_{10} = 45$$

Example IV

Find the sum of the following A.P

(i) $6 + 10 + 14 + \dots 50$

(ii) $10 + 7 + 4 + \dots -50$

(iii) $5 + 9 + 13 + \dots 101$

(iv) $1 + 1\frac{1}{4} + 1\frac{1}{2} + \dots 9\frac{3}{4}$

(v) $83 + 80 + 77 + \dots 5$

Solution

(i) $a = 6, d = 4$

$$U_n = a + (n - 1)d$$

$$50 = 6 + (n - 1)4$$

$$44 = 4(n - 1)$$

$$11 = n - 1$$

$$n = 12$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{12} = \frac{12}{2}(2 \times 6 + (12 - 1) \times 4)$$

$$= 6(12 + 44)$$

$$S_{12} = 6(56)$$

$$= 336$$

(ii) $10 + 7 + 4 + \dots - 50$

$$a = 10, \quad d = 7 - 10 = -3$$

$$n^{th} \text{ term} = a + (n - 1)d$$

$$-50 = 10 + (n - 1)(-3)$$

$$-60 = -3(n - 1)$$

$$20 = n - 1$$

$$n = 21$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{21} = \frac{21}{2}[2 \times 10 + (21 - 1) \times -3]$$

$$= \frac{21}{2}(20 + -60)$$

$$= \frac{21}{2}(-40)$$

$$= -420$$

(iii) $5 + 9 + 13 \dots 101$

$$a = 5, \quad d = 4,$$

$$n^{th} \text{ term} = a + (n - 1)d$$

$$101 = 5 + (n - 1)4$$

$$96 = 4(n - 1)$$

$$24 = n - 1$$

$$n = 25$$

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$\begin{aligned}
 S_{25} &= \frac{25}{2}(2 \times 5 + (25 - 1) \times 4) \\
 &= \frac{25}{2}(10 + 96) \\
 &= \frac{25}{2}(106) \\
 S_{25} &= 1325
 \end{aligned}$$

$$(iv) 1 + 1\frac{1}{4} + 1\frac{1}{2} \dots 9\frac{3}{4}$$

$$1 + \frac{5}{4} + \frac{3}{2} \dots \frac{39}{4}$$

$$a = 1, d = \frac{1}{4}$$

$$\frac{39}{4} = a + (n - 1)d$$

$$\frac{39}{4} = 1 + (n - 1)\frac{1}{4}$$

$$\frac{39}{4} = 1 + \frac{1}{4}n - \frac{1}{4}$$

$$\frac{39}{4} = \frac{3}{4} + \frac{1}{4}n$$

$$39 = 3 + n$$

$$n = 36$$

$$S_{36} = \frac{36}{2} \left(2 \times 1 + (36 - 1) \times \frac{1}{4} \right)$$

$$\begin{aligned}
 S_{36} &= 18(2 + 8.75) \\
 &= 193.5
 \end{aligned}$$

$$(v) 83 + 80 + 77 + \dots 5$$

$$a = 83, d = 80 - 83$$

$$d = -3$$

$$u_n = a + (n - 1)d$$

$$5 = 83 + (n - 1)(-3)$$

$$5 = 83 + -3n + 3$$

$$5 = 86 - 3n$$

$$3n = 81, n = 27$$

$$S_{27} = \frac{27}{2}(2 \times 83 + (27 - 1) \times -3)$$

$$= \frac{27}{2}(166 + -78)$$

$$= \frac{27}{2}(88)$$

$$= 1188$$

Example V

In an A.P the sum of the first 10 terms is 520 and the 7th terms doubles the 3rd term. Find the first term a and common difference (d).

Solution

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$S_{10} = 520$$

$$S_{10} = \frac{10}{2}(2a + (10 - 1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$520 = 5(2a + 9d)$$

$$104 = 2a + 9d \dots \dots \dots (1)$$

7th term

$$u_n = a + (n - 1)d$$

$$u_7 = a + 6d$$

3rd term

$$u_3 = a + 2d$$

$$u_7 = 2u_3$$

$$a + 6d = 2(a + 2d)$$

$$a + 6d = 2a + 4d$$

$$0 = a - 2d$$

$$a = 2d$$

Substituting eqn (2) in eqn (1)

$$104 = 2(2d) + 9d$$

$$104 = 4d + 9d$$

$$13d = 104$$

$$d = \frac{104}{13}$$

$$d = 8$$

$$a = 2d$$

$$a = 16$$

Example V

The first and last terms of an A.P with 25 terms are 29 and 179 respectively. Find the sum of the series and it's common difference

Solution

$$n = 25, a = 29, n^{\text{th}} \text{ term}$$

$$u_n = a + (n - 1)d$$

$$179 = 29 + (25 - 1)d$$

$$150 = 24d$$

$$d = 6.25$$

$$S_{25} = \frac{25}{2}(2 \times 29 + (25 - 1) \times 6.25)$$

$$S_n = 12.5(58 + 150)$$

$$= 2600$$

Example VI

The n th term of the series is $10-3n$. Find the first three terms of the series and the sum of the first 15 terms.

Solution

$$n^{\text{th}} \text{ term} = 10 - 3n$$

$$\text{first term } n = 1$$

$$\Rightarrow a = 10 - 3 \times 1$$

$$= 7$$

$$2^{\text{nd}} \text{ term } n = 2$$

$$= 10 - 3 \times 2$$

$$= 4$$

$$3^{\text{rd}} \text{ term} = 10 - 3 \times 3$$

$$= 1$$

$$7 + 4 + 1 + \dots$$

$$a = 7, \quad d = -3$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{15} = \frac{15}{2}(2 \times 7 + (15-1) \times -3)$$

$$= \frac{15}{2}(14 + -42)$$

$$= -210$$

Example VIII

Given that the first and third terms of an A.P are 13 and 25 respectively. Find the 100^{th} term and the sum of the first 15 terms

Solution

$$a = 13$$

$$a + 2d = 25$$

$$13 + 2d = 25$$

$$2d = 12$$

$$d = 6$$

$$n^{\text{th}} \text{ term} = a + (n-1)d$$

$$n^{\text{th}} \text{ term} = 13 + (n-1)6$$

$$100^{\text{th}} \text{ term} = 13 + (100-1) \times 6$$

$$u_{100} = 13 + 99 \times 6$$

$$= 607$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{15} = \frac{15}{2}(2 \times 13 + (100-1) \times 6)$$

$$S_n = 4650$$

Example VII

The second and seventh terms of an A.P are -5 and 10 respectively. Find the fifth term and the least number of terms that must be taken for their sum to exceed 200

Solution

$$a + d = -5 \dots\dots\dots (1)$$

$$a + 6d = 10 \dots\dots\dots (2)$$

$$\text{eqn (2)} - \text{eqn (1)}$$

$$5d = 15$$

$$d = 3$$

$$a + d = -5$$

$$a = -5 - d$$

$$a = -5 - 3$$

$$a = -8$$

$$u_n = a + (n-1)d$$

$$u_5 = -8 + (5-1) \times 3$$

$$u_5 = -8 + 12$$

$$u_5 = 4$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_n = \frac{n}{2}(2 \times -8 + (n-1)3)$$

$$S_n = \frac{n}{2}(-16 + 3n - 3)$$

$$S_n = \frac{n}{2}(-19 + 3n)$$

$$S_n > 200$$

$$S_n = \frac{n}{2}(-19 + 3n) > 200$$

$$-19n + 3n^2 > 400$$

$$3n^2 - 19n - 400 > 0$$

$$3\left(n^2 - \frac{19}{3}n\right) > 400$$

$$n^2 - \frac{19}{3}n > \frac{400}{3}$$

By completing squares;

$$n^2 - \frac{19n}{3} + \frac{361}{36} - \frac{361}{36} > \frac{400}{3}$$

$$\left(n - \frac{19}{6}\right)^2 > \frac{400}{3} + \frac{361}{36}$$

$$\left(n - \frac{19}{6}\right)^2 > \frac{5161}{36}$$

$$n - \frac{19}{6} > \sqrt{\frac{5161}{36}}$$

$$n > \frac{19}{6} + \sqrt{\frac{5161}{36}}$$

$$n > 15.1400167$$

$$n = 16$$

Example VIII

In an A.P the sum of the first 15 terms is 615 and the 13th term is six times the second term. Find the first three terms

Solution

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{15} = \frac{15}{2}(2a + (15-1)d) = 615$$

$$15(a + 7d) = 615$$

$$15a + 105d = 615$$

$$a + 7d = \frac{615}{15}$$

$$a + 7d = 41$$

$$\Rightarrow a = 41 - 7d \dots\dots\dots (1)$$

$$13^{th} \text{ term} = a + (13-1)d$$

$$u_{13} = a + 12d$$

$$2^{nd} \text{ term} = a + d$$

$$u_{13} = 6u_2$$

$$a + 12d = 6a + 6d$$

$$0 = 5a - 6d \dots\dots\dots (2)$$

Substituting Eqn (1) in Eqn (2)

$$\Rightarrow 0 = 5(41 - 7d) - 6d$$

$$0 = 205 - 35d - 6d$$

$$0 = 205 - 41d$$

$$d = 5$$

$$a = 41 - 7d$$

$$\Rightarrow a = 41 - 35$$

$$a = 6$$

The first three terms are: 6, 11, 17 ...

Example IX

In an A.P the sum of the first 2n terms is equal to the sum of the next n terms. If the first term is 12 and common difference is 3. Find the non-zero value of n.

Solution

In an A.P the sum of the first 2n terms

$$S_{2n} = \frac{2n}{2}(2 \times 12 + (2n-1)3)$$

$$= n(24 + 6n - 3)$$

$$= n(21 + 6n)$$

$$= 21n + 6n^2$$

The total of 2n terms and the next n terms is 3n terms.

$$S_{3n} = \frac{3n}{2}[(2 \times 12 + (3n-1)3)]$$

$$S_{3n} = \frac{3n}{2}[24 + 9n - 3]$$

$$= \frac{3n}{2}(21 + 9n)$$

$$= \frac{63n}{2} + \frac{27n^2}{2}$$

The sum of the next n terms after 2n terms is

$$\left(\frac{63n}{2} + \frac{27n^2}{2}\right) - (21n + 6n^2)$$

$$\frac{63n + 27n^2 - 42n - 12n^2}{2}$$

$$\frac{21n + 15n^2}{2}$$

Sum of first 2n terms = sum of the next n terms.

$$\Rightarrow 21n + 6n^2 = \frac{21n + 15n^2}{2}$$

$$42n + 12n^2 = 21n + 15n^2$$

$$0 = 3n^2 - 21n$$

$$0 = 3n(n-7)$$

$$n = 0 \quad n = 7$$

$$n \neq 0 \Rightarrow n = 7$$

Example X

The sum of the first n terms of a certain series is $3n^2 + n$. Show that the series is an A.P find the first term and common difference.

Solution

$$S_n = 3n^2 + n$$

$$S_{n-1} = 3(n-1)^2 + (n-1)$$

$$S_{n-1} = 3n^2 - 5n + 2$$

$n^{th} \text{ term} = S_n - S_{n-1}$ $= (3n^2 + n) - (3n^2 - 5n + 2)$
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$= 6n - 2$
 if $n = 1$ (1^{st} term)
 $a = 4$
 if $n = 2$ ($second$ term)
 $6 \times 2 - 2 = 10$
 If $n = 3$ ($third$ term)
 $6 \times 3 - 2 = 16$
 $first\ term = 4$
 Common difference = 6
 4, 10, 16, ...
 $\Rightarrow$ The series is an A.P

Example XI

The sum to n terms of a particular series is given by

$$S_n = 17n - 3n^2$$

- (a) Find an expression for the sum of $(n-1)$ terms
 (b) Find the expression for the n^{th} term of the series.
 Show that the series is arithmetic progression. Find the first term and common difference.

Solution:

$$\begin{aligned}
 S_n &= 17n - 3n^2 \\
 S_{(n-1)} &= 17(n-1) - 3(n-1)^2 \\
 &= 17n - 17 - 3(n^2 - 2n + 1) \\
 &= 17n - 17 - 3n^2 + 6n - 3 \\
 &= 23n - 3n^2 - 20 \\
 n^{th}\ term &= S_n - S_{n-1} \\
 &= (17n - 3n^2) - (23n - 3n^2 - 20) \\
 &= -6n + 20 \\
 &= 20 - 6n \\
 \text{For } n = 1. (\text{first term}) \\
 20 - 6 &= 14 \\
 \text{For } n = 2. (\text{second term}) \\
 20 - 6 \times 2 &= 8 \\
 \text{For } n = 3 (\text{third term}) \\
 20 - 6 \times 3 &= 2 \\
 \text{First term } a &= 14 \\
 \text{Common difference} \\
 d &= -6
 \end{aligned}$$

Example XII

The sum of the three terms in A.P is 30 and the sum of their squares is 398. Find the numbers.

Solution

$$\begin{aligned}
 a + (a + d) + a + 2d &= 30 \\
 3a + 3d &= 30 \\
 a + d &= 10 \dots\dots\dots (1) \\
 a^2 + (a + d)^2 + (a + 2d)^2 &= 398
 \end{aligned}$$

$$\begin{aligned}
 3a^2 + 6ad + 5d^2 &= 398 \dots\dots\dots (2) \\
 \text{from eqn (1)} \\
 a &= (10 - d) \\
 \text{Substituting } a &= 10 - d \text{ in Eqn (2);} \\
 \Rightarrow 3(10 - d)^2 + 6d(10 - d) + 5d^2 &= 398 \\
 3(100 - 20d + d^2) + 60d - 6d^2 + 5d^2 &= 398 \\
 300 - 60d + 3d^2 + 60d - d^2 &= 398 \\
 2d^2 &= 98 \\
 d^2 &= 49 \\
 d &= \pm 7 \\
 \text{If } d &= 7 \\
 a &= 3 \\
 3, 10, 17 \dots \\
 \text{If } d &= -7 \\
 a + d &= 10 \\
 a - 7 &= 10 \\
 a &= 17 \\
 17, 10, 3 \\
 \text{The numbers are } 3, 10 \text{ and } 17.
 \end{aligned}$$

Example XIII

Show that the sum to 20 terms of the series

$$\log a + \log ab + \log ab^2 + \log ab^3 + \dots$$

can be written in the form $\log a^x b^y$ and find x and y .

Solution

$$\begin{aligned}
 d &= \log ab - \log a \\
 &= \log b \\
 S_n &= \frac{n}{2}(2a + (n-1)d) \\
 S_{20} &= \frac{20}{2}[(2 \log a + (20-1) \log b)] \\
 &= 10[(\log a^2 + 19 \log b)] \\
 &= 10[(\log a^2 + \log b^{19})] \\
 &= 10[\log a^2 b^{19}] \\
 &= \log (a^2 b^{19})^{10} \\
 &= \log a^{20} b^{190} \\
 &= \log a^x b^y \\
 x &= 20, \quad y = 190
 \end{aligned}$$

GEOMETRIC PROGRESSION

A geometric progression G.P is a sequence where each new terms after the first is obtained by multiplying the preceding term by a constant r , called a common ratio. If

the first term of the sequence is 'a' then the geometric progression is

$$a + ar + ar^2 + ar^3 + \dots ar^{n-1}$$

$a = \text{first term}, r = \text{common ratio}$

$ar^{n-1} = n^{\text{th}} \text{ term.}$

Examples of geometric progression are:

(i) $2 + 6 + 18 + 54 + \dots$

(ii) $1 + 2 + 4 + 8 + \dots 128 + 256$

(iii) $27 - 9 + 3 - 1 + \dots \frac{1}{27} - \frac{1}{81}$

From the above examples

$$2 + 6 + 18 + 54 + \dots$$

First term ($a = 2$)

$$\text{common ratio} = \frac{6}{2} = \frac{18}{6} = \frac{54}{18} = 3$$

For $1 + 2 + 4 + 8 + \dots 128 + 256$

$$a = 1, \text{ common ratio } (r) = \frac{2}{1} = \frac{4}{2} = \frac{8}{4} = 2$$

For $27 - 9 + 3 - 1 + \dots \frac{1}{27} - \frac{1}{81}$

First term $a = 27$

$$\text{common ratio } r = \frac{-9}{27} = \frac{3}{-9} = -\frac{1}{3}$$

Example I

Which of the following series are geometric progression write down the common ratios of those that are:

(i) $3 + 9 + 27 + 81 \dots$

(ii) $-1 + 2 - 4 + 8 \dots$

(iii) $1 + 1.1 + 1.21 + 1.331 \dots$

(iv) $1 + 1/4 + 1/16 + 1/64 + \dots$

(v) $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{36} \dots$

Solution

(i) $3 + 9 + 27 + 81 \dots$

$$r_1 = \frac{9}{3} = 3$$

$$r_2 = \frac{27}{9} = 3$$

$$r_3 = \frac{81}{27} = 3$$

$$r_1 = r_2 = r_3 = \text{common ratio} = r = 3$$

$\Rightarrow 3 + 9 + 27 + 81 \dots$ is a geometric progression

(ii) $-1 + 2 - 4 + 8$

$$r_1 = \frac{2}{-1} = -2$$

$$r_2 = \frac{-4}{-2} = -2$$

$$r_3 = \frac{8}{-4} = -2$$

$$r_1 = r_2 = r_3 = r = -2$$

Common ratio

$\Rightarrow -1 + 2 - 4 + 8$ is a geometric progression with a common ratio $r = -2$

(iii) $1 + 1.1 + 1.21 + 1.331 \dots$

$$r_1 = \frac{1.1}{1} = 1.1$$

$$r_2 = \frac{1.21}{1.1} = 1.1$$

$$r_3 = \frac{1.331}{1.21} = 1.1$$

It is a geometric progression with a common ratio $r = 1.1$

(iv) $1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} \dots$

$$r_1 = \left(\frac{\frac{1}{4}}{1} \right) = \frac{1}{4}$$

$$r_2 = \left(\frac{\frac{\frac{1}{16}}{\frac{1}{4}}}{\frac{1}{4}} \right) = \frac{1}{16} \times 4 = \frac{1}{4}$$

$$r_3 = \left(\frac{\frac{\frac{1}{64}}{\frac{1}{16}}}{\frac{1}{16}} \right) = \frac{1}{64} \times 16 = \frac{1}{4}$$

It is a geometric progression with a common ratio $r = \frac{1}{4}$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{36} + \dots$$

$$r_1 = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{6} \times 2 = \frac{1}{3}$$

$$r_2 = \frac{\frac{1}{12}}{\frac{1}{6}} = \frac{6}{12} = \frac{1}{2}$$

$$r_3 = \frac{\frac{1}{36}}{\frac{1}{12}} = \frac{12}{36} = \frac{1}{3}$$

$$r_1 \neq r_2$$

$\Rightarrow \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{36} \dots$ is not a geometric progression because the common ratio is not the same.

Example II

State the common ratio and the next two term of the following geometric progression (G.P)

(i) $1 - 2 + 4 - 8 + \dots$

$$(ii) \frac{1}{4} + \frac{1}{2} + 1 + \dots$$

$$(iii) 200 + 50 + 12\frac{1}{2} + \dots$$

$$(iv) 162 + 54 + 18 + \dots$$

Solution

Let the next two terms be x and y

$$1 + -2 + 4 + -8 + x + y + \dots$$

$$r = -\frac{2}{1} = -2 = \text{common ratio}$$

$$\Rightarrow \frac{x}{-8} = -2$$

$$x = 16$$

$$\text{And } \frac{y}{x} = -2$$

$$\frac{y}{16} = -2$$

$$y = -32$$

$\Rightarrow$ The next two numbers of the sequence

$1 + -2 + 4 + -8 + \dots$ are 16 and -32

$$(ii) \frac{1}{4} + \frac{1}{2} + 1 + \dots$$

$$r = \frac{\frac{1}{2}}{\frac{1}{4}} = \frac{1}{2} \times \frac{4}{1} = 2$$

Let the next numbers of the sequence be m and n .

$$\frac{1}{4} + \frac{1}{2} + 1 + m + n + \dots$$

$$\frac{m}{\frac{1}{4}} = 2 \Rightarrow m = 2$$

$$\frac{n}{m} = 2, \quad n = 2m$$

$$n = 4$$

$\Rightarrow$ The next two numbers of the sequence

$$\frac{1}{4} + \frac{1}{2} + 1 + \dots \text{ are } 2 \text{ and } 4$$

$$(iii) 200 + -50 + 12\frac{1}{2} + \dots$$

$$r = -\frac{50}{200} = -\frac{1}{4}$$

Let the next two numbers of the series be m and n

$$\Rightarrow 200 + -50 + 12\frac{1}{2} + m + n + \dots$$

$$\frac{m}{12.5} = -\frac{1}{4}$$

$$4m = -12.5$$

$$m = -3.125, \quad m = \frac{-25}{8}$$

$$m = -3\frac{1}{8}$$

$$\frac{n}{m} = -\frac{1}{4}$$

$$n = -\frac{1}{4}m$$

$$n = -\frac{1}{4} \left(\frac{-25}{8} \right)$$

$\Rightarrow$ The next two terms of the series

$$200 + -50 + 12\frac{1}{2} + \dots \text{ are } -\frac{25}{8} \text{ and } \frac{25}{32}$$

$$(iv) 162 + 54 + 18 + \dots$$

Let the next terms of the sequence be m and n .

$$162 + 54 + 18 + m + n + \dots$$

$$\frac{m}{18} = \frac{1}{3}$$

$$3m = 18$$

$$m = 6$$

$$\frac{n}{m} = \frac{1}{3}$$

$$\frac{n}{6} = \frac{1}{3}$$

$$3n = 6$$

$$n = 2$$

$\Rightarrow$ The next two terms of the sequence

$$162 + 54 + 18 + m + n + \dots \text{ are } 6 \text{ and } 2$$

Example III

Write down the terms indicated in each of the following progression

$$(i) \frac{1}{2} + 1 + 2 + \dots 8^{th}$$

$$(ii) 200 - 50 + 2\frac{1}{2} + \dots 5^{th}$$

$$(iii) 162 + 54 + 18 + \dots 6^{th}$$

$$(iv) -\frac{4}{9} - \frac{2}{3} - 1 + \dots 7^{th}$$

Solution

The n^{th} of a geometric progression is given by;

$$U_n = ar^{n-1}$$

$$\text{For } \frac{1}{2} + 1 + 2 + \dots 8^{th}$$

$$a = \frac{1}{2} \quad r = \frac{1}{\frac{1}{2}} = 2$$

$$\Rightarrow U_n = \frac{1}{2}(2^{n-1})$$

For the 8^{th} term, $n = 8$

$$\Rightarrow U_8 = \frac{1}{2}(2^7) = 64$$

(ii) $200 - 50 + 12.5 + \dots$

The n^{th} term $U_n = ar^{n-1}$

$$a = 200 \quad r = -\frac{1}{4}$$

$$U_n = 200 \left(-\frac{1}{4}\right)^{n-1}$$

For the 5th term

$$U_5 = 200 \left(-\frac{1}{4}\right)^{5-1}$$

$$U_5 = \frac{25}{32}$$

$\Rightarrow$ The 5th term is $\frac{25}{32}$

Example IV

$162 + 54 + 18 + \dots$ 6th term

$$U_n = ar^{n-1}$$

$$r = \frac{54}{162} = \frac{1}{3}$$

$$a = 162$$

$$U_n = 162 \left(\frac{1}{3}\right)^{n-1}$$

$$6^{\text{th}} \text{ term} = 162 \left(\frac{1}{3}\right)^{6-1}$$

$$= \frac{2}{3}$$

$\Rightarrow$ The 6th term is $\frac{2}{3}$

$$-\frac{4}{9} - \frac{2}{3} - 1 + \dots$$
 7th

$$a = -\frac{4}{9}, \quad r = -\left(\frac{\frac{2}{3}}{-\frac{4}{9}}\right)$$

$$r = \frac{-2}{3} \times \frac{9}{-4} = \frac{3}{2}$$

$$U_n = ar^{n-1}$$

$$= \frac{-4}{9} \left(\frac{3}{2}\right)^{n-1}$$

For the 7th term, $n = 7$

$$U_7 = \frac{-4}{9} \left(\frac{3}{2}\right)^{7-1}$$

$$= \frac{-4}{9} \left(\frac{729}{64}\right) = \frac{-81}{16}$$

$\Rightarrow$ The 7th term is $\frac{-81}{16}$

Example VI

How many term are in a geometric progression

$$2 + 4 + 8 + \dots + 128$$

Solution:

The n^{th} term is given by $U_n = ar^{n-1}$

$$a = 2, \quad r = \frac{4}{2} = 2$$

$$\Rightarrow 128 = 2 \times (2)^{n-1}$$

$$128 = 2^n$$

$$2^7 = 2^n$$

$$n = 7$$

Example VII

The first term of a G.P with positive common ratio is 80.

If the sum of the first three term is 185, find the common ratio.

Solution

$a = 80$ (first term).

$$a + ar + ar^2 = 185$$

$$a(1 + r + r^2) = 185$$

$$80(1 + r + r^2) = 185$$

$$1 + r + r^2 = \frac{185}{80}$$

$$1 + r + r^2 = \frac{37}{16}$$

$$16 + 16r + 16r^2 = 37$$

$$16r^2 + 16r - 21 = 0$$

$$(4r + 7)(4r - 3) = 0$$

$$r = \frac{3}{4}, \quad r = -\frac{7}{4}$$

Example VIII

In a G.P the second term exceeds the first by 20 and the fourth exceeds the second by 15. Find the two possible values of the first term

Solution

$$2^{\text{nd}} \text{ term} = ar$$

$$1^{\text{st}} \text{ term} = a$$

$$a + 20 = ar$$

$$20 = ar - a \dots\dots\dots (1)$$

$$4^{th} term = ar^3$$

$$second\ term = ar$$

$$ar + 15 = ar^3$$

$$15 = ar^3 - ar \dots\dots\dots (2)$$

$$Eqn\ (1) \div Eqn(2)$$

$$\frac{20}{15} = \frac{a(r-1)}{ar(r^2-1)}$$

$$\frac{4}{3} = \frac{r-1}{r(r^2-1)}$$

$$\frac{4}{3} = \frac{r-1}{r^3-r}$$

$$4r^3 - 4r = 3r - 3$$

$$4r^3 - 4r - 3r + 3 = 0$$

$$4r^3 - 7r + 3 = 0$$

$$r = 1, \quad r = -\frac{3}{2}, \quad r = \frac{1}{2}$$

$$\text{If } r = \frac{1}{2}, \quad 20 = ar - a$$

$$\Rightarrow 20 = \frac{1}{2}a - a$$

$$20 = -\frac{a}{2}$$

$$a = -40$$

$$\text{When } r = -\frac{3}{2},$$

$$20 = -\frac{3}{2}a - a$$

$$20 = -\frac{5}{2}a$$

$$-\frac{40}{5} = a$$

$$a = -8$$

When $r = 1$, a is not defined.

Example IX

The second and fifth term in a G.P are 405 and -120 respectively. Find the seventh term.

Solution

$$ar = 405 \dots\dots\dots (1)$$

$$ar^4 = -120 \dots\dots\dots (2)$$

$$Eqn\ (2) \div Eqn\ (1)$$

$$r^3 = \frac{-120}{405}$$

$$r^3 = \frac{-8}{27}$$

$$r = -\frac{2}{3}$$

$$a\left(-\frac{2}{3}\right) = 405$$

$$a = -607.5$$

$$ar^6 = (-607.5)\left(-\frac{2}{3}\right)^6$$

$$= \frac{-160}{3}$$

Example IX

The second, fourth and eighth terms of an A.P are in geometric progression, the sum of the third and fifth terms is 20. Find the first four terms of the progression.

Solution

$$second\ term = a + d$$

$$4^{th}\ term = a + 3d$$

$$8^{th}\ term = a + 7d$$

$$(a + d), \quad a + 3d, \quad a + 7d$$

$$\frac{a + 3d}{a + d} = \frac{a + 7d}{a + 3d}$$

$$(a + 3d)^2 = (a + d)(a + 7d)$$

$$a^2 + 6ad + 9d^2 = a^2 + 7ad + ad + 7d^2$$

$$2d^2 - 2ad = 0$$

$$2d(d - a) = 0$$

$$d = 0 \quad \text{but } d \neq 0$$

$$d = a \dots\dots\dots (1)$$

$$a + 2d + a + 4d = 20$$

$$2a + 6d = 20$$

$$a + 3d = 10 \dots\dots\dots (2)$$

Substitute Eqn (1) in Eqn (2)

$$a + 3a = 10$$

$$a = 2.5$$

$$2.5, 5, 7.5, 10 \dots$$

Sum of a Geometric Progression

Let the sum of a geometric progression be S_n

$$S_n = a + ar + ar^2 + \dots ar^{n-2} + ar^{n-1} \dots\dots\dots (1)$$

multiplying equation (1) by r

$$rS_n = ar + ar^2 + ar^3 + \dots ar^{n-1} + ar^n \dots\dots\dots (2)$$

$$Eqn\ (1) - Eqn(2)$$

$$\Rightarrow S_n - rS_n = a - ar^n$$

$$S_n(1 - r) = a(1 - r^n)$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad \text{for } r < 1$$

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ for } r > 1$$

Example I

Find the sum of six terms of the progression

$$2 + 6 + 18 + 54 + \dots$$

Solution

$$a = 2, \quad r = \frac{6}{2} = 3$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_6 = \frac{2(3^6 - 1)}{3 - 1}$$

$$= 728$$

Example II

Find the sum of the first 20 terms of a G.P with first term 3 and common ratio 1.5

Solution

$$a = 3, \quad r = 1.5$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{20} = \frac{3(1.5^{20} - 1)}{1.5 - 1}$$

$$S_{20} = 19945.54038$$

Example III

Find the sum of the first six terms of a G.P

$$100 + 10 + 1 + \dots$$

Solution

$$a = 100, \quad r = \frac{10}{100} = \frac{1}{10}$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_6 = \frac{100 \left(1 - \left(\frac{1}{10} \right)^6 \right)}{1 - \frac{1}{10}}$$

$$S_6 = 111.111$$

Example IV

Find the sum of the following G.P Progression

a) $\frac{1}{4} + \frac{1}{2} + \dots + 64$

b) $1000 + 200 + \dots + 0.32$

c) $\frac{1}{4} - \frac{1}{2} + \dots + 64$

d) $2 - 3 + \dots + 22\frac{25}{32}$

Solution

$$\frac{1}{4} + \frac{1}{2} + \dots + 64$$

$$a = \frac{1}{4}, \quad r = \left(\frac{\frac{1}{2}}{\frac{1}{4}} \right) = \frac{1}{2} \times 4 = 2$$

$$ar^{n-1} = u_n$$

$$\Rightarrow \frac{1}{4} (2)^{n-1} = 64$$

$$4 \times \frac{1}{4} (2)^{n-1} = 64 \times 4$$

$$2^{n-1} = 256$$

$$2^{n-1} = 2^8$$

$$n - 1 = 8$$

$$n = 9$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_9 = \frac{\frac{1}{4}(2^9 - 1)}{2 - 1}$$

$$S_9 = \frac{1}{4}(511)$$

$$S_9 = 127.75$$

(ii) $1000 + 200 + \dots + 0.32$

$$a = 1000, \quad r = \frac{200}{1000} = 0.2$$

$$ar^{n-1} = u_n$$

$$1000 (0.2)^{n-1} = 0.32$$

$$(0.2)^{n-1} = 0.00032$$

$$(0.2)^{n-1} = (0.2)^5$$

$$n - 1 = 5$$

$$n = 6$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_6 = \frac{\frac{1}{4}(1 - 0.2^6)}{1 - 0.2}$$

$$S_6 = \frac{1}{4} \left(\frac{0.999936}{0.8} \right)$$

$$= 0.31248$$

(c) $\frac{1}{4} - \frac{1}{2} + \dots + 64$

$$a = \frac{1}{4} \quad r = -\frac{\frac{1}{2}}{\frac{1}{4}} = -\frac{1}{2} \times 4 = -2$$

$$ar^{(n-1)} = 64$$

$$\frac{1}{4}(-2)^{n-1} = 64$$

$$(-2)^{n-1} = 64 \times 4$$

$$-2^{n-1} = 256$$

$$(-2)^{n-1} = (-2)^8$$

$$n - 1 = 8$$

$$n = 9$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_9 = \frac{\frac{1}{4}(1 - (-2)^9)}{1 - -2}$$

$$S_9 = \frac{1}{4} \left(\frac{1 - -512}{3} \right)$$

$$S_9 = 42.75$$

$$(d) \quad 2 + -3 + \dots 22 \frac{25}{32}$$

$$a = 2, \quad r = \frac{-3}{2}$$

$$ar^{n-1} = 22 \frac{25}{32}$$

$$2 \left(-\frac{3}{2} \right)^{n-1} = \frac{729}{32}$$

$$\left(\frac{-3}{2} \right)^{n-1} = \frac{729}{32} \div 2$$

$$\left(\frac{-3}{2} \right)^{n-1} = \frac{729}{64}$$

$$\left(-\frac{3}{2} \right)^{n-1} = \left(-\frac{3}{2} \right)^6$$

$$n - 1 = 6$$

$$n = 7$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_7 = \frac{2 \left(1 - \left(-\frac{3}{2} \right)^7 \right)}{1 - \frac{-3}{2}}$$

$$S_7 = 14.46875$$

Example V

Find the sum of the first n terms of the G.P

$$\frac{1}{12} + \frac{1}{4} + \frac{3}{4} + \dots$$

how many terms of the series are needed to reach a sum greater than 100

Solution

$$\frac{1}{12} + \frac{1}{4} + \frac{3}{4} + \dots$$

$$a = \frac{1}{12}, \quad r = \frac{\frac{1}{4}}{\frac{1}{12}} = \frac{1}{4} \times 12$$

$$r = 3$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{\frac{1}{12}(3^n - 1)}{3 - 1}$$

$$S_n = \frac{1}{12} \left(\frac{3^n - 1}{2} \right)$$

$$S_n = \frac{3^n - 1}{24}$$

$$S_n > 100$$

$$\frac{3^n - 1}{24} > 100$$

$$3^n - 1 > 2400$$

$$3^n > 2401$$

$$\log_{10} 3^n > \log_{10} 2401$$

$$n > \frac{\log_{10} 2401}{\log_{10} 3}$$

$$n > 7.08497$$

$$n = 8$$

Example VI

The sum of the first seven term of a G.P is 7 and the sum of next seven terms is 896. Find the common ratio of the progression. If the k^{th} term is the first term of a G.P which is greater than 1, find k .

Solution

$$\text{sum of the first seven terms} = 7$$

$$\text{sum of the next seven term} = 896$$

$$\begin{aligned} \text{sum of the 14 terms} &= 7 + 896 \\ &= 903 \end{aligned}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{14} = \frac{a(r^{14} - 1)}{r - 1}$$

$$\frac{a(r^{14} - 1)}{r - 1} = 903$$

$$a(r^{14} - 1) = 903(r - 1) \dots \dots \dots (1)$$

$$S_7 = 7$$

$$\frac{a(r^7 - 1)}{r - 1} = 7$$

$$a(r^7 - 1) = 7(r - 1) \dots \dots \dots (2)$$

$$\text{Eqn (1)} \div \text{Eqn (2)}$$

$$\frac{r^{14} - 1}{r^7 - 1} = 129$$

$$\text{let } r^7 = P$$

$$\frac{P^2 - 1}{P - 1} = 129$$

$$\frac{(p+1)(p-1)}{p-1} = 129$$

$$p + 1 = 129$$

$$p = 128$$

$$\text{But } p = r^7$$

$$r^7 = 128$$

$$\Rightarrow r = 2$$

Example VII

A G.P has first term 16 and common ratio $\frac{3}{4}$. If the sum of the first n terms is greater than 60. Find the least possible values of n .

Solution

$$a = 16, \quad r = \frac{3}{4}$$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$\frac{16 \left(1 - \left(\frac{3}{4} \right)^n \right)}{1 - \frac{3}{4}} > 60$$

$$1 - \left(\frac{3}{4} \right)^n > \frac{15}{16}$$

$$\frac{1}{16} > \left(\frac{3}{4} \right)^n$$

$$\log_{10} \left(\frac{1}{16} \right) > \log_{10} \left(\frac{3}{4} \right)^n$$

$$\log_{10} \left(\frac{1}{16} \right) > n \log_{10} \frac{3}{4}$$

$$\frac{\log_{10} \left(\frac{1}{16} \right)}{\log_{10} \left(\frac{3}{4} \right)} < n$$

$$9.638 < n$$

$$\Rightarrow n = 10$$

Sum to Infinity of A G.P

Consider the general geometric progression (G.P)

$$a + ar + ar^2 + \dots + ar^{n-1}$$

The sum of the n terms is denoted by

$$S_n = \frac{a(1 - r^n)}{1 - r}, \text{ for } |r| < 1$$

Since n cannot be negative it implies that as n increases r^n decreases for $|r| < 1$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_n = \frac{a}{1 - r} - \frac{a r^n}{1 - r}$$

But as n tends to a big positive value ($n \rightarrow \infty$),

$$\frac{a r^n}{1 - r} \approx 0$$

$$\Rightarrow S_\infty = \frac{a}{1 - r} \text{ (sum to infinity of a G.P)}$$

We say that the G.P with a common ratio, r and first term (a) converges when $-1 < r < 1$ and the limit of the sum is $\frac{a}{1 - r}$. There is no limit of a geometric progression whose common ratio lies outside the range. $|r| \leq 1$

Example I

Find the sum to infinity of a G.P with first term 3 and common ratio $\frac{1}{2}$

Solution

$$S_\infty = \frac{a}{(1 - r)}$$

$$S_\infty = \frac{3}{1 - \frac{1}{2}}$$

$$S_\infty = \frac{3}{\frac{1}{2}}$$

$$S_\infty = 6$$

Example II

Find the sum to infinity of

$$1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots$$

Solution

$$a = 1, \quad r = \frac{1}{3}$$

$$S_\infty = \frac{a}{1 - r}$$

$$S_{\infty} = \frac{1}{1 - \frac{1}{3}}$$

$$= \frac{1}{\frac{2}{3}} = \frac{3}{2}$$

Example III (UNEB Question)

Find how many terms of the series

$$1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots$$

must be taken so that the sum will differ from the sum to infinity by less than 10^{-6}

Solution

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_n = \frac{1 \left(1 - \left(\frac{1}{5} \right)^n \right)}{1 - \frac{1}{5}}$$

$$S_n = \frac{1 - \left(\frac{1}{5} \right)^n}{\frac{4}{5}}$$

$$S_n = \frac{5}{4} \left(1 - \left(\frac{1}{5} \right)^n \right)$$

$$S_{\infty} = \frac{1}{1 - r}$$

$$S_{\infty} = \frac{1}{1 - \frac{1}{5}} = \frac{5}{4}$$

$$S_{\infty} - S_n < 10^{-6}$$

$$\frac{5}{4} - \left(\frac{5}{4} \left(1 - \left(\frac{1}{5} \right)^n \right) \right) < 10^{-6}$$

$$\frac{5}{4} \left(\frac{1}{5} \right)^n < 10^{-6}$$

$$\left(\frac{1}{5} \right)^n < \frac{10^{-6} \times 4}{5}$$

$$\left(\frac{1}{5} \right)^n < 8 \times 10^{-7}$$

$$(0.2)^n < 8 \times 10^{-7}$$

$$n(\log_{10} 0.2) < \log_{10} 8 \times 10^{-7}$$

$$n > \frac{\log_{10} 8 \times 10^{-7}}{\log_{10} 0.2}$$

(Sign changes because the denominator is negative)

$$n > 8.7227$$

$$n > 8.7227$$

$$n = 9$$

Example IV

The sum to infinity of a G.P is twice the sum of the first two terms. Find the possible values of the common ratio.

Solution

$$S_{\infty} = \frac{a}{1 - r}$$

$$S = a + ar$$

$$2(a + ar) = \frac{a}{1 - r}$$

$$2(1 + r) = \frac{1}{1 - r}$$

$$2(1 - r^2) = 1$$

$$1 - r^2 = \frac{1}{2}$$

$$1 - \frac{1}{2} = r^2$$

$$r^2 = \frac{1}{2}$$

$$r = \pm \frac{1}{\sqrt{2}}$$

Example V

The first and fourth terms of a geometric series are 135 and -40 respectively. Find its common ratio and the sum to infinity.

Solution

$$a = 135$$

$$ar^3 = -40$$

$$135r^3 = -40$$

$$r^3 = -\frac{40}{135}$$

$$r = \frac{-2}{3}$$

$$S_{\infty} = \frac{a}{1 - r}$$

$$S_{\infty} = \frac{135}{1 - \frac{-2}{3}}$$

$$S_{\infty} = 81$$

Example VI

The sum of the first n terms of a geometric progression is $8 - 2^{3-2n}$. Find the first term, its common ratio and its sum to infinity.

Solution

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$8 - 2^{3-2n} = \frac{a(1-r^n)}{1-r}$$

$$2^3 - 2^{3-2n} = \frac{a(1-r^n)}{1-r}$$

$$2^3(1 - (2^{-2})^n) = \frac{a(1-r^n)}{1-r}$$

$$2^3 \left(1 - \left(\frac{1}{4}\right)^n\right) = \frac{a(1-r^n)}{1-r}$$

$$\frac{1}{4} = r \text{ by Comparison}$$

$$\frac{a}{1-r} = 2^3,$$

The sum to infinity = 8

$$\frac{a}{3/4} = 2^3$$

$$\Rightarrow a = 8 \times \frac{3}{4}$$

$$a = 6$$

$$S_\infty = \frac{a}{1-r}$$

$$= \frac{6}{1-\frac{1}{4}}$$

$$= \frac{6}{3/4} = 8$$

Example (UNEB Question)

In an arithmetic progression $u_1 + u_2 + u_3 + \dots, u_4 = 15$ and $u_{16} = -3$. Find the greatest integer N such that $u_N \geq 0$. Determine the sum of the first N terms of the progression.

Solution

Let $a = 1^{\text{st}}$ term of the A.P. and
 $d =$ common difference of the A.P.

The n^{th} term of the A.; $u_n = a + (n-1)d$

$\Rightarrow 4^{\text{th}}$ term, $u_4 = a + 3d$

But $u_4 = 15$

$\Rightarrow a + 3d = 15 \dots\dots\dots (i)$

The 16^{th} term $u_{16} = a + 15d$

But $u_{16} = -3$

$\Rightarrow a + 15d = -3 \dots\dots\dots (ii)$

Eqn (i) – Eqn (ii)

$-12d = 18$

$d = -1.5$

From eqn (i)

$a + 3(-1.5) = 15$

$a = 15 + 4.5$

$a = 19.5$

Substituting for a and d in $u_n = a + (n-1)d$

$\Rightarrow u_n = 19.5 + (n-1) \times -1.5$

$= 19.5 - 1.5(n-1)$

Now for $u_n \geq 0$

$\Rightarrow 19.5 - 1.5(n-1) \geq 0$

$19.5 - 1.5n + 1.5 \geq 0$

$21 - 1.5n \geq 0$

$21 \geq 1.5n$

$14 \geq n$

Hence $n \leq 14$

The greatest integer, N is 14

Sum of n terms of the A.P. is given by

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_n = \frac{14}{2} [2 \times 19.5 + (14-1) \times -1.5]$$

$$= 7 [39 + (13 \times -1.5)]$$

$$= 7 (39 - 19.5)$$

$$= 136.5$$

Example (UNEB Question)

Show that $\ln 2^r$, $r = 1, 2, 3$ is an arithmetic progression.

ii) Find the sum of the first 10 terms of the progression.

iii) Determine the least value of m for which the sum of the first $2m$ terms exceeds 883.7.

Solution

a) i) For $r = 1$,

$\ln 2^1 = \ln 2$

For $r = 2$

$$\ln 2^2 = 2 \ln 2$$

Common difference = $2 \ln 2 - \ln 2 = \ln 2$

For $r = 3$

$$\ln 2^3 = 3 \ln 2$$

Common difference = $3 \ln 2 - 2 \ln 2 = \ln 2$

Since the difference between any two consecutive terms is the same i.e. $\ln 2$, Therefore the progression is an arithmetic progression.

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

ii) For an A.P.,

Where $n = 10$, $a = \ln 2$ and $d = \ln 2$

$$\Rightarrow S_{10} = \frac{10}{2} [2 \ln 2 + (10-1) \ln 2]$$

$$= 5 [2 \ln 2 + (10-1) \ln 2]$$

$$= 55 \ln 2$$

$$= 38.1231 \text{ (4 dp)}$$

iii) Give $a = \ln 2$

$$d = \ln 2$$

$$n = 2m$$

$$\Rightarrow S_{2m} = \frac{2m}{2} [2 \ln 2 + (2m-1) \ln 2]$$

$$= m[(2+2m-1) \ln 2]$$

$$= m[(1+2m) \ln 2]$$

$$\text{For } S_{2m} > 883.7,$$

$$\Rightarrow m(1+2m) \ln 2 > 883.7$$

$$m + 2m^2 > \frac{883.7}{\ln 2}$$

$$2m^2 + m > 1274.9$$

$$2m^2 + m - 1274.9 > 0$$

$$m > \frac{-1 \pm \sqrt{1+8 \times 1274.9}}{4}$$

$$m > \frac{-1 \pm \sqrt{10200.2}}{4}$$

$$m > \frac{-1 \pm 100.996}{4}$$

$$m > 24.999$$

Hence the least value of m is 25.

Example (UNEB Question)

The first term of an arithmetic progression (A.P) is 73 and the 9th is 25. Determine

- The common difference of the A.P.
- The number of terms that must be added to give a sum of 96.

b) A geometric progression (G.P.) and an arithmetic progression (A.P) have the same first term. The sums of their first, second and third terms are 6, 10.5 and 18 respectively. Calculate the sum of their fifth terms.

Solution

Let $a = 1^{\text{st}}$ term and

$d = \text{common difference of an A.P}$

Now 1^{st} term = 73 and

The 9th term = 25

But 9th term = $a + 8d$

$$\Rightarrow a + 8d = 25$$

$$73 + 8d = 25$$

$$8d = 25 - 73$$

$$8d = 48$$

$$d = -6$$

Hence the common difference of the A.P is -6

ii) Let $n = \text{number of terms required}$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

But $S_n = 96$

$$\Rightarrow \frac{n}{2} [2a + (n-1)d] = 96$$

Substituting for $a = 73$ and $d = -6$

$$\frac{n}{2} [146 + (n-1) \times -6] = 96$$

$$\frac{n}{2} [146 - 6(n-1)] = 96$$

$$n[73 - 3(n-1)] = 96$$

$$73n - 3n^2 + 3n = 96$$

$$3n^2 - 76n + 96 = 0$$

$$n = \frac{76 \pm \sqrt{76^2 - 4 \times 3 \times 96}}{2 \times 3}$$

$$= \frac{76 \pm \sqrt{5776 - 1152}}{6}$$

$$= \frac{76 \pm 68}{6}$$

$$n = \frac{76 + 68}{6}$$

$$= 24$$

Hence the terms that must be added to give a sum of 96 are 24

b) Let $a = 1^{\text{st}}$ term of both A.P. and G.P.

$d = \text{common difference of A.P. and}$

$r = \text{common difference of a G.P.}$

Now for an A.P:

$$1^{\text{st}} \text{ term} = a$$

$$2^{\text{nd}} \text{ term} = a + d$$

$$3^{\text{rd}} \text{ term} = a + 2d$$

$$4^{\text{th}} \text{ term} = a + 3d$$

$$5^{\text{th}} \text{ term} = a + 4d$$

Also for a G.P.:

$$1^{\text{st}} \text{ term} = a$$

$$2^{\text{nd}} \text{ term} = ar$$

$$3^{\text{rd}} \text{ term} = ar^2$$

$$4^{\text{th}} \text{ term} = ar^3$$

$$5^{\text{th}} \text{ term} = ar^4$$

Sum of their 1st terms = $2a$

$$\Rightarrow 2a = 6$$

$$a = 3$$

Sum of their second terms = $a + d + ar$

$$\Rightarrow a + d + ar = 10.5$$

$$3 + d + 3r = 10.5$$

$$d + 3r = 7.5 \dots \dots \dots (i)$$

Sum of their third terms = $a + 2d + ar^2$

$$\Rightarrow a + 2d + ar^2 = 18$$

$$3 + 2d + 3r^2 = 18$$

$$2d + 3r^2 = 15$$

Making d the subject from Eqn (ii)

$$d = 7.5 - 3r$$

Substitute for d into Eqn (i)

$$2(7.5 - 3r) + 3r^2 = 15$$

$$15 - 6r + 3r^2 = 15$$

$$3r^2 - 6r = 0$$

$$r = 2$$

$$d = 7.5 - 3 \times 2 = 1.5$$

The sum of their 5th terms = $a + 4d + ar^4$

$$= 3 + 4 \times 1.5 + 3 \times 2^4$$

$$= 3 + 6 + 48$$

$$= 57$$

Example (UNEB Question)

a) The n^{th} term of a series is $U_n = a3^n + bn + c$. given that $U_1 = 4$, $U_2 = 13$ and $U_3 = 46$, find the values of a , b and c .

Solution

a) Given $U_n = a3^n + bn + c$

Substituting for $n = 1, 2, 3...$

For $n = 1$

$$\Rightarrow 3a + b + c = 4 \dots\dots\dots(i)$$

For $n = 2$,

$$\Rightarrow 9a + 2b + c = 13 \dots\dots\dots(ii)$$

For $n = 3$

$$\Rightarrow 27a + 3b + c = 46 \dots\dots\dots(iii)$$

Eqn (ii) – Eqn (i)

$$6a + b = 9 \dots\dots\dots(iv)$$

Eqn (iii) – Eqn (ii)

$$18a + b = 33 \dots\dots\dots(v)$$

Eqn (v) – Eqn (iv)

$$12a = 24$$

$$a = 2$$

Substitute for a in Eqn (iv)

$$12 + b = 9$$

$$b = -3$$

Substituting for b in Eqn (i),

$$6 - 3 + c = 4$$

$$c = 1$$

$$\therefore a = 2, b = -3 \text{ and } c = 1$$

Example (UNEB Question)

9.(a) The tenth term of an arithmetic progression (A.P) is 29 and the fifteenth term is 44. Find the value of the common difference and the first term. Hence find the sum of the first 60 terms.

b) A cable 10 m long is divided into ten pieces whose lengths are in a geometric progression. The length of the

longest piece is 8 times the length of the shortest piece.

Calculate to the nearest centimeter the length of the third piece.

Solution:

(a) The n^{th} term of A.P. is given by

$$U_n = a + (n - 1)d$$

Given $U_{10} = 29$,

$$\Rightarrow a + 9d = 29 \dots\dots\dots(i)$$

Given $U_{15} = a + 14d$

$$\Rightarrow a + 14d = 44 \dots\dots\dots(ii)$$

Eqn (i) – Eqn (ii);

$$5d = 15$$

$$d = 3$$

Substituting for d in Eqn (i)

$$a + 27 = 29$$

$$a = 2$$

The sum of the first n terms of A.P. is given by

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{60} = \frac{60}{2} [2(2) + 59(3)]$$

$$= 30[4 + 177]$$

$$= 5430$$

b) Given length of cable = 10m

Number of divisions = 10

Here, length of divisions makes a G.P.

By considering the first to be the shortest and the 10th to be the longest,

Let 1st term = a

10th term = ar^9

3rd term = ar^2

$$ar^9 = 8a$$

$$r^9 = 8$$

$$r = 8^{\frac{1}{9}} = 2^{\frac{1}{3}}$$

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right)$$

$$10 = a \left(\frac{8^{\frac{10}{9}} - 1}{8^{\frac{1}{9}} - 1} \right)$$

$$10 = a \left(\frac{2^{\frac{10}{3}} - 1}{2^{\frac{1}{3}} - 1} \right)$$

$$10 = a \left(\frac{9.0794}{0.2599} \right)$$

$$a = 0.2863 \quad (4d.p.)$$

$$\begin{aligned}\text{Length of the 3rd piece, } U_3 &= ar^2 \\ U_3 &= 0.2862 \times 2^{\frac{1}{2}} \\ &= 0.4544\text{m} \\ &\simeq 45 \text{ cm}\end{aligned}$$

Application of Geometric Progression

UNEB Question

A credit society gives out a compound interest of 4.5% per annum. Mugagga deposited 300,000 at the beginning of each year. How much money will he have at the end of the 4th year in no withdrawal is made between this period.

Solution

$$\begin{aligned}A &= A_1 + A_2 + \dots A_n \\ A_1 &= 300,000 \left(1 + \frac{4.5}{100}\right) \\ A_2 &= 300,000 \left(1 + \frac{4.5}{100}\right)^2 \\ A_3 &= 300,000 \left(1 + \frac{4.5}{100}\right)^3 \\ A_4 &= 300,000 \left(1 + \frac{4.5}{100}\right)^4\end{aligned}$$

$$A = 300,000(1 + 0.045) + 300,000(1 + 0.045)^2 + \dots 300,000(1 + 0.045)^4$$

$$A = 300,000(1.045 + 1.045^2 + \dots 1.045^4)$$

$$A = 300,000a \frac{(r^n - 1)}{r - 1}$$

$$a = 1.045$$

$$r = \frac{(1.045)^2}{1.045} = 1.045$$

$$A = \frac{300,000 \times 1.045(1.045^4 - 1)}{1.045 - 1}$$

$$A = 1341212.918$$

Example II (UNEB Question)

A Finance society gives out a compound interest of 8% per annum. Moses deposited £100 into a saving account at the beginning of each year. How much will his saving be worth after ten years?

Solution:

$$A = A_1 + A_2 + A_3 + \dots A_{10}$$

$$\begin{aligned}A &= 100 \left(1 + \frac{8}{100}\right) + 100 \left(1 + \frac{8}{100}\right)^2 \\ &\quad + \dots 100 \left(1 + \frac{8}{100}\right)^{10} \\ A &= 100(1.08 + 1.08^2 + \dots 1.08^{10}) \\ A &= 100 \frac{a(r^n - 1)}{r - 1} \\ A &= \frac{100 \times 1.08(1.08^{10} - 1)}{1.08 - 1} \\ A &= 1564.55 \\ &= \text{€}1564.5548746\end{aligned}$$

Example III

5 millions are invested each year at a rate of 15% interest. In how many years will it accumulate to more than 50 millions.

$$\begin{aligned}A &= A_1 + A_2 + A_3 + \dots A_n \\ A &= 5(1.15 + 1.15^2 + \dots 1.15^n) \\ a &= 1.15 \\ r &= \frac{1.15^2}{1.15} = 1.15\end{aligned}$$

$$A = \frac{5a(r^n - 1)}{r - 1}$$

$$A = \frac{5 \times 1.15(1.15^n - 1)}{1.15 - 1}$$

$$\Rightarrow \frac{5 \times 1.15(1.15^n - 1)}{1.15 - 1} > 50.$$

$$1.15^n > \frac{53}{23}$$

$$\log_{10} 1.15^n > \log_{10} \left(\frac{53}{23}\right).$$

$$n \log_{10} 1.15 > \left(\log_{10} \frac{53}{23}\right).$$

$$n \log_{10} 1.15 > \left(\log_{10} \frac{53}{23}\right)$$

$$n > \frac{\log_{10} \frac{53}{23}}{\log_{10} 1.15}$$

$$n > 5.97$$

$$n = 6.$$

Example IV

A man pays a premium of £100 at the beginning of every year to insurance company on understanding that at the end of fifteen years he can receive back the premiums which he has paid with 5% compound interest. What should he receive? Give your answer correct to three significant figure.

Solution

$$A = 100 \left(1 + \frac{5}{100}\right)^1 + 100 \left(1 + \frac{5}{100}\right)^2 + \cdots 100 \left(1 + \frac{5}{100}\right)^{15}$$

$$A = 100[(1.05 + 1.05^2 + \cdots 1.05^{15})]$$

$$A = \frac{100 a (r^n - 1)}{r - 1}$$

$$A = \frac{100 \times 1.05(1.05^{15} - 1)}{1.05 - 1}$$

$$A = 2265.74917.$$

$$A = \text{£}2270 \text{ (to 3 significant figure)}$$

Proof by induction of summation of other series

Mathematical induction is a method of proof in which a statement is proved for one step in a process and it's shown that if the statement holds for that step, it holds for the next. Proof by induction is not a direct proof because the method doesn't produce the expression it's self but it can only be used to prove that a given expression is a required sum.

Mathematic proof by induction involves 3 steps

Step 1: (initial step) is to prove the given statement for all natural numbers.

Step 2: (induction step) prove the given statement for any natural number implies the given statement for the next natural numbers.

Step 3: (conclusion step).

If you have just covered induction with your teacher and you are feeling uneasy about the whole thing. If you are anything like me, you are having the same thought that I

had. Can't you prove anything is true. If you assume it to be true in the first place, what is left is to prove.

If you have already proved isn't that cheating any how

Well.....

First step don't get mad at your teacher.

Induction makes a perfect sense to him and he honestly thinks he has explained it clearly. When I was still at school like you I had a good math teacher so I had good lessons of induction from my teacher.

But still I didn't trust induction so I tried to pick up where my teacher had left me and fill in some gaps.

Before the lesson ended. I asked my teacher, 'excuse me teacher'

Excuse me teacher, 'excuse me teacher, excuse me teacher. Mathematical proof by induction is confusing me because I cannot seem to reason with myself as to how to go about getting to the solution and also asked why is it called mathematical proof by induction not substitution induction'.

Example 1

Prove by induction that

$$1 + 2 + 3 + \cdots n = \frac{1}{2}n(n + 1)$$

Solution

$$1 + 2 + 3 + \cdots n = \frac{1}{2}n(n + 1)$$

$$\text{For } n = 1$$

$$1 = \frac{1}{2}(1)(2)$$

$$1 = 1.$$

It is true for $n = 1$

Assume the result holds for some general value of n say $n = k$.

$$1 + 2 + 3 + \cdots K = \frac{1}{2}k(k + 1)$$

It must also be true for the next value of n i.e

$$n = k + 1$$

Adding the next term, $n = k + 1$

$$\begin{aligned} \Rightarrow 1 + 2 + 3 + \cdots + k + k + 1 &= \frac{1}{2}k(k + 1) + k + 1 \\ &= (k + 1)\left[\frac{1}{2}k + 1\right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2}(k+1)[k+2] \\
&= \frac{1}{2}(k+1)(k+2)
\end{aligned}$$

$$\Rightarrow 1 + 2 + 3 + \dots + k + k + 1 = \frac{1}{2}(k+1)(k+2)$$

For the next integer

$$\begin{aligned}
n &= k + 1 = 2. \\
k + 1 &= 2. \\
k &= 1. \\
\Rightarrow 1 + 2 &= \frac{1}{2}(1+1)(1+2) \\
3 &= \frac{1}{2}(2)(3) \\
3 &= 3.
\end{aligned}$$

Since it's true for $n = 1$, $n = 2$ and so on hence it's true for all positive integral values of n .

Example II

Prove by induction that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1).$$

Solution

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

For $n = 1$

$$1 = \frac{1}{6}(1)(2)(3).$$

$$1 = 1$$

It's true for $n = 1$.

Assume the result holds for the general value of n say $n=k$, it must be true for the next value of n i.e

$$n = k + 1$$

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{6}k(k+1)(2k+1)$$

Adding the next term $n = k + 1$

$$\begin{aligned}
1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \frac{1}{6}(k)(k+1)(2k+1) + (k+1)^2 \\
&= \frac{1}{6}(k+1)[2k^2 + k + 6(k+1)] \\
&= \frac{1}{6}(k+1)[2k^2 + 7k + 6] \\
&= \frac{1}{6}(k+1)[(k+2)(2k+3)]
\end{aligned}$$

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6}(k+1)(2k+3)$$

For the next integer

$$n = k + 1 = 2$$

$$k = 1.$$

$$1^2 + 2^2 = \frac{1}{6}(2)(3)(5)$$

$$5 = 5$$

Since it is true for $n = 1$ and $n = 2$ and so on, and so on hence it's true for all positive integral values of n .

Example III

Prove by induction that

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

Solution

For $n = 1$.

$$a = a \frac{(1-r)}{1-r}$$

$$a = a$$

It's true for $n = 1$

Assume the result holds for the general value of n say $n = k$

$$a + ar + ar^2 + \dots + ar^{k-1} = a \frac{(1-r^k)}{1-r}$$

It must be true for the next integer

$$n = k + 1.$$

Adding the next term

$$n = k + 1$$

$$\begin{aligned}
a + ar + ar^2 + \dots + ar^{k-1} + ar^k &= a \frac{(1-r^k)}{1-r} + ar^k \\
&= \frac{a(1-r^k) + ar^k(1-r)}{1-r} \\
&= \frac{a - ar^k + ar^k - ar^{k+1}}{1-r} \\
&= \frac{a - ar^{k+1}}{1-r} \\
&= \frac{a(1-r^{k+1})}{1-r}
\end{aligned}$$

$$a + ar + ar^2 + \dots + ar^{k-1} + ar^k = \frac{a(1-r^{k+1})}{1-r}$$

For the next integer

$$n = k + 1 = 2.$$

$$k = 1.$$

$$a + ar = \frac{a(1-r^2)}{1-r}$$

$$a + ar = \frac{a(1+r)(1-r)}{1-r}$$

$$a + ar = a(1+r)$$

$$a + ar = a + ar$$

Since it's true for $n = 1$, $n = 2$, $n = k$ so on hence it's true.

Example IV

Prove by induction that $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$.

Solution

$$\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{n(n+1)}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

For $n = 1$

$$\frac{1}{2} = \frac{1}{2}.$$

It's true for $n = 1$.

Assume the result holds for general value of n , say $n = k$.

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

It must be true for the next integer $n = k + 1$

adding the next term $n = k + 1$

$$\begin{aligned} \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2) + 1}{(k+1)(k+2)} \\ &= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\ &= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{k+1}{k+2} \end{aligned}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

For the next integer, $n = k + 1 = 2, k = 1$.

$$\frac{1}{2} + \frac{1}{6} = \frac{2}{3}.$$

$$\frac{3+1}{6} = \frac{2}{3}.$$

$$\frac{4}{6} = \frac{2}{3}.$$

Since it is true for $n = 1, n = 2$ and so on hence it's true for all positive integral value of n .

Example V

Prove by induction that $\sum_{m=1}^n m^3 = \frac{1}{4}n^2(n+1)^2$

Solution

$$\sum_{m=1}^n m^3 = 1^3 + 2^3 + 3^3 + \dots + n^3$$

$$\Rightarrow 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$$

for $n = 1$

$$1^3 = \frac{1}{4}(1^2)(2)^2$$

$$1 = 1.$$

It's true for $n = 1$.

Assume the result holds for the general value of

$n = k$

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}k^2(k+1)^2$$

It must be true for next integer $n = k + 1$

Adding the next term $n = k + 1$

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 \\ &= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \\ &= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)] \\ &= \frac{1}{4}(k+1)^2(k^2 + 4k + 4) \\ &= \frac{1}{4}(k+1)^2(k+2)^2 \end{aligned}$$

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

$$1^3 + 2^3 = \frac{1}{4}(2)^2(3^2)$$

$$9 = 9$$

Since it true for $n = 1, n = 2$ and so on, it is true for all positive integral value of n .

Example VI

Prove by induction that

$$1 \times 2 + 2 \times 3 + \dots + n(n+1) = \frac{1}{3}n(n+1)(n+2)$$

Solution

$$(1 \times 2) + (2 \times 3) + \dots + n(n+1) = \frac{1}{3}n(n+1)(n+2)$$

For $n = 1$

$$1 \times 2 = \frac{1}{3}(1)(2)(3)$$

$$2 = 2.$$

Its true result holds for general value of n say
 $n = k$.

$$(1 \times 2) + (2 \times 3) + \cdots k(k+1) = \frac{1}{3}k(k+1)(k+2)$$

It must be true for the next integer $n = k + 1$

$$(1 \times 2) + (2 \times 3) + \cdots k(k+1) + (k+1)(k+2)$$

$$= \frac{1}{3}k(k+1)(k+2) + (k+1)(k+2)$$

$$= \frac{1}{3}(k+1)(k+2)[k+3].$$

$$= \frac{1}{3}(k+1)(k+2)(k+3)$$

$$1 \times 2 + 2 \times 3 + \cdots k(k+1) + (k+1)(k+2)$$

$$= \frac{1}{3}(k+1)(k+2)(k+3)$$

For the next integer, $n = k + 1 = 2, k = 1$

$$1 \times 2 + 2 \times 3 = \frac{1}{3}(2)(3)(4).$$

$$8 = 8$$

Since it is true for $n = 1, n = 2$ and so on hence it's true for all positive integral values of n .

Example VII

Prove by induction that

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Solution

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

For $n = 1$.

$$\frac{d}{dx}(x) = 1.$$

Assume the result holds for the general value of n say
 $n = k$.

$$\frac{d}{dx}(x^k) = kx^{k-1}$$

It must be true for the next integer $n = k + 1$

$$\frac{d}{dx}(x^{k+1}) = (k+1)x^k.$$

For the next integer, $n = k + 1 = 2, k = 1$

$$\frac{d}{dx}(x^2) = 2x$$

Since it is true for $n = 1, n = 2$, and so on, it is true for positive integral values of n .

Example VIII

Prove by induction.

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(4n^2 - 1)$$

Solution

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(4n^2 - 1)$$

For $n = 1$

$$1^2 = \frac{1}{3}(1)(3)$$

$$1 = 1$$

It is true for $n = 1$.

Assume the results holds for the general value of $n = k$.

$$1^2 + 3^2 + 5^2 + \cdots \dots (2k-1)^2 = \frac{1}{3}k(4k^2 - 1)$$

It must be true for the next integer $n = k + 1$

Adding the next term $n = k + 1$

$$1^2 + 3^2 + 5^2 + \cdots (2k-1)^2 + (2k+1)^2$$

$$= \frac{1}{3}k(4k^2 - 1) + (2k+1)^2$$

$$= \frac{1}{3}k(2k+1)(2k-1) + (2k+1)^2$$

$$= \frac{1}{3}(2k+1)[2k^2 - k + 3(2k+1)]$$

$$= \frac{1}{3}(2k+1)[(2k^2 + 5k + 3)]$$

$$= \frac{1}{3}(2k+3)(2k+1)(k+1)$$

$$1^2 + 3^2 + \cdots (2k-1)^2 + (2k+1)^2$$

$$= \frac{1}{3}(k+1)(2k+1)(2k+3)$$

For the next integer

$$n = k + 1 = 2.$$

$$k = 1$$

$$1^2 + 3^2 = \frac{1}{3}(2)(3)(5)$$

$$10 = 10$$

Since it is true for $n = 1, n = 2$ and so on hence it is true for all positive integral values of n .

Example IX

Prove by induction that $6^n - 1$ is divisible by 5 for all positive integral value of n

Solution

$$\text{let } a_n = \frac{6^n - 1}{5}$$

$$\text{for } n = 1$$

$$a_1 = \frac{6 - 1}{5} = 1.$$

It's true for $n = 1$

Assume it holds for the general value of n say $n = k$.

$$a_k = \frac{6^k - 1}{5}$$

$$5a_k = 6^k - 1 \dots\dots\dots (i)$$

It must be true for the next integer $n = k + 1$

$$a_{k+1} = \frac{6^{k+1} - 1}{5}$$

$$5a_{k+1} = 6^{k+1} - 1 \dots\dots\dots (ii)$$

$eqn(ii) - eqn(i)$

$$\begin{aligned} 5a_{k+1} - 5a_k &= (6^{k+1} - 1) - (6^k - 1) \\ &= 6^{k+1} - 6^k \\ &= 6^k(6^1 - 1) \\ &= 5(6^k) \end{aligned}$$

For the next integer

$$n = k + 1 = 2$$

$$k = 1.$$

$$5a_2 - 5a_1 = 5 \times 6$$

$$5a_2 - 5 \times 1 = 30.$$

$$5a_2 = 35.$$

$$a_2 = 7$$

Since it is true for $n = 1$, $n = 2$ and so on, it is true for all the positive integral values of n .

Example X

Prove by induction that $4^n - 1$ is divisible by 3 for all positive integral value of n .

Solution

$$a_n = \frac{4^n - 1}{3}$$

for $n = 1$

$$a_1 = \frac{4^1 - 1}{3} = 1$$

It is true for $n = 1$

Assume the result holds for general value of n say $n = k$

$$a_k = \frac{4^k - 1}{3}$$

$$3a_k = 4^k - 1 \dots\dots\dots (i)$$

It must be true for the next integer $n = k + 1$

$$a_{k+1} = \frac{4^{k+1} - 1}{3}$$

$$3a_{k+1} = 4^{k+1} - 1 \dots\dots\dots (ii)$$

$eqn(ii) - eqn(i)$

$$\begin{aligned} 3a_{k+1} - 3a_k &= (4^{k+1} - 1) - (4^k - 1) \\ &= 4^{k+1} - 4^k \\ &= 4^k(4 - 1) \end{aligned}$$

$$= 3(4^k).$$

For the next integer, $n = k + 1 = 2, k = 1$.

$$3a_2 - 3a_1 = 3(4^1).$$

$$3a_2 - 3 = 12$$

$$3a_2 = 15$$

$$a_2 = 5.$$

Since it's true for $n = 1, n = 2, n = k$ and so on it's true for all positive integral values of n .

Example

Prove by induction that $8^n - 7n + 6$ is divisible by 7 for all positive values of n .

Solution

$$a_n = \frac{8^n - 7n + 6}{7}$$

For $n = 1$

$$a_1 = \frac{8 - 7 + 6}{7} = 1$$

It's true for $n = 1$ for the general value of n say $n = k$.

$$a_k = \frac{8^k - 7k + 6}{7}$$

$$7a_k = 8^k - 7k + 6 \dots\dots\dots (1)$$

It must be true for the next integer

$n = k + 1$

$$a_{k+1} = \frac{8^{k+1} - 7(k+1) + 6}{7}$$

$$7a_{k+1} = 8^{k+1} - 7k - 7 + 6.$$

$$7a_{k+1} = 8^{k+1} - 7k - 1 \dots\dots\dots (2)$$

$eqn(2) - eqn(1)$.

$$\begin{aligned} 7a_{k+1} - 7a_k &= (8^{k+1} - 7k - 1) - (8^k - 7k + 6) \\ &= 8^{k+1} - 8^k - 7 \\ &= 8^k \cdot 8 - 8^k - 7. \\ &= 8^k(8 - 1) - 7 \\ &= 7 \cdot 8^k - 7 \\ &= 7(8^k - 1) \\ 7(a_{k+1} - a_k) &= 7(8^k - 1) \end{aligned}$$

For the next integer

$$n = k + 1 = 2, k = 1$$

$$7(a_2 - a_1) = 7(8^1 - 1).$$

$$a_2 - a_1 = 7$$

$$a_2 = 7 + a_1$$

$$a_2 = 7 + 1$$

$$a_2 = 8$$

Since it is true for $n = 1, n = 2$ and so on it's true for all positive integral values of n .

Example (UNEB Question)

Prove by induction that $2^n + 3^{2n-3}$ is always divisible by 7 for $n \geq 2$

Solution

$$a_n = \frac{2^n + 3^{2n-3}}{7}$$

for $n = 2$

$$a_2 = \frac{2^2 + 3}{7} = 1$$

it's true for $n = 1$.

Assume the result holds for the general value of n say $n = k$

$$a_k = \frac{2^k + 3^{2k-3}}{7}$$

$$7a_k = 2^k + 3^{2k-3} \dots \dots \dots (1)$$

For the next integer, $n = k + 1$

$$a_{k+1} = \frac{2^{k+1} + 3^{2k-1}}{7}$$

$$7a_{k+1} = 2^{k+1} + 3^{2k-1} \dots \dots \dots (2)$$

Eqn (2) – Eqn (1)

$$7a_{k+1} - 7a_k = (2^{k+1} + 3^{2k-1}) - (2^k + 3^{2k-3})$$

$$= 2^k \cdot 2 + 3^{2k-3} \cdot 3^2 - 2^k - 3^{2k-3}$$

$$= 2(2^k) - 2^k + 3^2(3^{2k-3}) - 3^{2k-3}$$

$$= 2^k(2 - 1) + 3^{2k-3}(9 - 1)$$

$$= 2^k + 8(3^{2k-3})$$

$$= 2^k + 3^{2k-3} + 7 \cdot 3^{2k-3}$$

$$= 7a_k + 7 \cdot 3^{2k-3}$$

$$= 7(a_k + 3^{2k-3})$$

$$7(a_{k+1} - a_k) = 7(a_k + 3^{2k-3})$$

For the next integer $n = k + 1 = 3, k = 2$

$$7(a_3 - a_2) = 7(a_2 + 3)$$

$$a_3 - a_2 = a_2 + 3$$

$$a_3 = 2a_2 + 3$$

$$a_3 = 2 + 3$$

$$a_3 = 5$$

Since it is true for $n = 2, n = 3, n = k$ and so on it's true for all positive of n .

Example XI

Prove by induction that $8^n - 6^n$ is always divisible by 7 for all even integers of n .

Solution

$$a_n = \frac{8^n - 6^n}{7}$$

$$\text{For } n = 2, a_2 = \frac{8^2 - 6^2}{7} = \frac{64 - 36}{7} = 4$$

Assume the result holds for the general even integer $n = k$

$$a_k = \frac{8^k - 6^k}{7}$$

$$7a_k = 8^k - 6^k \dots \dots \dots (1)$$

$$\text{For the next even integer, } a_{k+2} = \frac{8^{k+2} - 6^{k+2}}{7}$$

$$7a_{k+2} = 8^{k+2} - 6^{k+2} \dots \dots \dots (2)$$

Eqn (2) – Eqn (1)

$$7(a_{k+2} - a_k) = 8^{k+2} - 6^{k+2} - 8^k + 6^k$$

$$7(a_{k+2} - a_k) = 8(8^2 - 1) - 6^k(6^2 - 1)$$

$$= 8^k(63) - 35(6^k)$$

$$a_{k+2} - a_k = 9(8^k) - 5(6^k)$$

$$a_{k+2} = a_k + 9(8^k) - 5(6^k)$$

For the next integer $n = k + 2 = 4, k = 2$

$$a_4 = a_2 + 9(8^2) - 5(6^2)$$

$$a_4 = 4 + 396$$

$$a_4 = 400$$

$$\text{From } a_n = \frac{8^n - 6^n}{7}, a_4 = \frac{8^4 - 6^4}{7} = 400$$

Since it is true for $n = 2, n = 4$, and so on, hence it is true for all positive even integers.

Revision Exercise

- Which of the following series are arithmetical progressions? Write down the common difference of those that are.
 - $7 + 8\frac{1}{2} + 10 + 11\frac{1}{2} + \dots$
 - $-2 - 5 - 8 - 11 + \dots$
 - $1 + 1.1 + 1.2 + 1.3 + \dots$
 - $1 + 1.1 + 1.1 + 1.11 + \dots$
 - $\frac{1}{2}, + \frac{5}{6} + \frac{7}{6} + \frac{3}{2} + \dots$
 - $1^2 + 2^2 + 3^2 + 4^2 + \dots$
 - $n + 2n + 3n + 4n + \dots$
 - $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$
- Write down the first term and common difference of each of the following arithmetic progressions.
 - $8 + 11 + 14 + 17 + \dots$
 - $23 + 25 + 27 + 29 + \dots$
 - $19 + 16 + 13 + 10 + \dots$
 - $13\frac{1}{2} + 15 + 16\frac{1}{2} + 18 + \dots$

3. Find the sums of the following arithmetic progressions
 - (a) $4 + 11 + \dots$ to 16^{th} term
 - (b) $3 + 8\frac{1}{2} + \dots$ to 20^{th} term
 - (c) $19 + 13 + \dots$ to 10^{th} term.
 - (d) $-9 - 1 + \dots$ to 8^{th} term.
4. Write down the terms indicated in each of the following Arithmetic progressions.
 - (a) $3 + 11 + \dots$, 10^{th} and 19^{th}
 - (b) $8 + 5 + \dots$, 15^{th} , 31^{st} term.
 - (c) $\frac{1}{4} + \frac{7}{8} + \dots$ 12^{th} , n^{th}
 - (d) $50 + 48 + \dots$, 100^{th} , n^{th}
5. Find the 18^{th} term of a series that has an n^{th} term given by $(2 + 3n)$
6. Find the sum of the arithmetic progression $-7 - 3 + 1 + \dots$ from the 7^{th} to the 13^{th} term inclusive.
7. Find the number of terms in the following arithmetic progressions:
 - (a) $2 + 4 + 6 + \dots + 46$
 - (b) $50 + 47 + 44 + \dots + 14$
 - (c) $2 + 4 + \dots + 4n$
 - (d) $a + (a + d) + \dots + l$
8. Find the 31^{st} term of a series that has an n^{th} term given by $\frac{1}{3}(10 + 2n)$
9. Find the sum of all odd numbers between 0 and 500 which are divisible by 7.
10. The 2^{nd} term of an arithmetic progression is 15 and the 15^{th} is 21. Find the common difference, the first term and the sum of the first 10 terms.
11. Find the 5^{th} and 7^{th} terms of a series that has an n^{th} term given by $(-1)^n(2n + 1)$
12. Show that the sum $1 + 3 + 5 + \dots + (2n - 1)$ is always a perfect square.
13. The 4^{th} term of an arithmetic progression is 18, and the common difference is -5. Find the 1^{st} term and the sum of the 1^{st} 16 terms.
14. Find an expression for the n^{th} term of each of the following arithmetic progression and use your answer to write down the 100^{th} term of each series.
 - (a) $5 + 8 + 11 + 14 + \dots$
 - (b) $5 + 2 + -1 - 4 + \dots$
 - (c) $12\frac{1}{2} + 16 + 19\frac{1}{2} + 23 + \dots$
15. A piece of string of length 5m is cut into n pieces in such way that the lengths of the piece are in an arithmetic progression. If the lengths of the longest and shortest pieces are 1m and 25 cm respectively, calculate n .
16. Find the difference between the sums of the first 10 terms of the A.P whose first terms are 12 and 8 and whose common differences are respectively 2 and 3.
17. Find the sum of each of the following arithmetic progressions:
 - (a) $2 + 4 + 6 + 8 + 10 + \dots$ 146
 - (b) $100 + 95 + 90 + 85 + 80 + \dots + 20$
 - (c) $4 + 10 + 16 + 22 + 28 + \dots + 334$
 - (d) $5\frac{1}{4} + 4\frac{1}{2} + 3\frac{3}{4} + \dots + -3$
18. The 10^{th} term of an arithmetic progression is 10 and the sum of the first 10 terms is -35. Find the first term and the common difference of the progression.
19. The first term of an arithmetic progression is -12, and the last term is 40. If the sum is 196, find the number of terms and the common difference.
20. In an arithmetic progression, $u_5 = -0.5$ and $S_7 = 21$. Find a , d , and u_6 .
21. The sum of the first four terms of an arithmetic progression is twice the 5^{th} term. Show that the common difference is equal to the first term.
22. Find the sum of the even number divisible by 3 lying between 400 and 500.
23. The sum of the first ten terms of an arithmetic progression is 120 and the sum of the first 20 terms is 840. Find the sum of the first 30 terms.
24. Find the arithmetic mean of:
 - (a) 3 and 27
 - (b) 3 and -27
 - (c) $\frac{1}{3}$ and $\frac{1}{27}$
 - (d) $\log 3$ and $\log 27$
25. Show that the sum of the integers from 1 to n is $\frac{1}{2}n(n + 1)$.
26. An arithmetic progression has a common difference, d . If the sum to 20 terms is 25 times the first term, find in terms of d , the sum to 30 terms.
27. Three numbers in an arithmetic progression have sum 33 and product 1232. Find the numbers.
28. Show that the sum of the first n terms of the arithmetic progression with first term a and common difference d is $\frac{1}{2}n(2a + (n - 1)d)$.
29. In an arithmetic progression, $a = -61$ and $d = 4$. Find the least value of n such that $S_n > 0$.
30. An arithmetic progression has first term -5 and common difference 1.5. Find the greatest number of terms the arithmetic progression can have given that the sum of the terms does not exceed 450.

Answers

1. (a) $1\frac{1}{2}$ (b) -3 (c) 0.1 (d) $\frac{1}{3}$ (g) n
2. (a) 8.3 (b) 23, 2 (c) 19, -3 (d) $13\frac{1}{2}$, $1\frac{1}{2}$
3. (a) 904 (b) 1188 (c) 88 (d) $193\frac{1}{2}$

4. (a) 75, 147 (b) -34, -82 (c) $7\frac{1}{8}, \frac{1}{8}(5n-3)$
 (d) -148, $(52-2n)$
 5. 56 6. 1512
 7. (a) 23 (b) 13 (c) $2n$ (d) $\frac{(l-a)}{(d+1)}$
 8. 24 9. 9072 10. 2, 13, 220
 11. 13, -15 13. 33, -72
 14. (a) $2+3n$, 302
 (b) $8-3n$, -292
 (c) $\frac{1}{2}(18+7n)$, 359
 15. 8 16. 5
 17. (a) 5402 (b) 1000
 (c) 9464 (d) $13\frac{1}{2}$
 18. -17, 3 19. 14, 4
 20. $13\frac{1}{2}$, $-3\frac{1}{2}$, $-14\frac{1}{2}$
 22. 7650 23. 2160
 24. (a) 15 (b) -15 (c) $\frac{5}{27}$ (d) $\log 9$
 26. 1575d 27. 8, 11, 14
 29. 32 30. 28.

Revision Exercise two

- For each of the following G.Ps, state the common ratio and the next two terms.
 - $4+20+100+500+\dots$
 - $24+12+6+\dots$
 - $45+15+5+\dots$
- Write down the terms indicated in each of the following G.Ps. Do not simplify your answer.
 - $5+10+\dots, 11^{\text{th}}, 20^{\text{th}}$
 - $10+25+\dots, 7^{\text{th}}, 19^{\text{th}}$
 - $\frac{2}{3}+\frac{3}{4}+\dots, 12^{\text{th}}, n^{\text{th}}$
 - $3-2+\dots, 8^{\text{th}}, n^{\text{th}}$
- Find the sum of the following G.Ps
 - $100+10+\dots$ to 7 terms
 - $1-\frac{1}{3}+\dots$ to 6 terms
 - $3-6+\dots$ to n terms
 - $a^p+a^{p+3}+a^{p+6}+\dots$ to k terms
- Using the formula $S_n = \frac{a(r^n-1)}{(r-1)}$, find S_5 and S_6 for the G.P 18, -9, $4\frac{1}{2}$, ... and hence deduce the value of U_6 .
- Find the number of terms in the following geometric progressions:
 - $81+27+9+\dots+1/27$
 - $0.03+0.06+0.12+\dots-1\frac{11}{16}$
 - $\frac{8}{81}-\frac{4}{27}+\frac{2}{9}-\dots-1\frac{11}{16}$
 - $5+10+20+\dots+5\times 2^n$
- (e) $a+ar+ar^2+\dots+ar^{n-1}$
- Find the distinct numbers p and q such that $p, q, 10$ are in arithmetic progression and $p, q, 10$ are in geometric progression.
- Find the value of the common ratio of the G.P that has a third term equal to 6 and 8th term equal to 1458.
- The third term of a geometric progression is 10 and the 6th term is 80. Find the common ratio, the first term and the sum of the first six terms.
- Find the geometric mean of:
 - 3 and 27
 - $\frac{1}{3}$ and $\frac{1}{27}$
 - 10^3 and 10^{27}
- In a geometric progression, the 7th term equals 8 and the 9th term equals 18. Find the possible values of the common ratio.
- The third term of a G.P is 2 and the fifth is 18. Find two possible values of the common ratio and the second term in each case.
- Given that the geometric mean of the numbers $4x-3$ and $9x+4$ is $6x-1$, find the value of x .
- Find the sum of the first ten terms of a G.P that has a sixth term $\frac{32}{33}$ and a seventh term of $1\frac{21}{33}$.
- The three numbers $n-1, n, n+3$ are consecutive terms of a geometric progression. Find n and the term after $n+3$.
- If the sum of the first two terms of a G.P is 162 and the sum of its first four terms is 180, find the sum of the first 6 terms, Find also the possible values of the sixth term..
- The geometric mean of two numbers a and b ($b > a$) is equal to four fifth of the arithmetic mean of the two numbers. If $a = 6$, find the value of b .
- A man starts saving on 1st April. He saves 1p the first day, 2p the second, 4p the third and so on. If he managed to keep on saving under this system until the end of the month (30 days), how much would he have saved?
- The sum of the first six terms of a G.P is nine times the sum of the first 3 terms. Find the common ratio.
- Prove that the arithmetic mean of two different numbers exceeds the geometric mean of the same two numbers.
- Show that the sum of the series $4+12+36+108+\dots$ to 20 terms is greater than 3×10^9 .
- The sum of $(n+12)$ terms of the G.P $2+4+8+\dots$ is twice the sum of n terms of the G.P $2+12+48+\dots$. Calculate the value of n .
- A geometric series has first term 5 and common ratio 3. Find the least number of terms the series can have if its series exceeds 2000.

23. Find the ratio of the sum of the first 10 terms of the series $\log x + \log x^2 + \log x^4 + \log x^8 + \dots$ to the first term.
24. A geometric series has first term 35 and common ratio 2^x . State the set of values of x for which the series is convergent. Find the value of x for which the sum to infinity of the series is 40.
25. The sum of the first two terms of a G.P is 9 and the sum to infinity of the G.P is 25. If the G.P has a positive common ratio r , find r and the first term.
26. The 2nd, 4th, 8th terms of an A.P are in geometric progression and the sum of the third and fifth terms is 20. Find the first four terms of the progression.
27. For each of the following geometric series, find the range of values of x for which the sum to infinity of the series exists.
- (a) $x + x^2 + x^3 + x^4 + \dots$
- (b) $1 + \frac{x}{3} + \frac{x^2}{9} + \frac{x^3}{27} + \dots$
- (c) $1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots$
- (d) $\left(x + \frac{1}{2}\right) + \left(x + \frac{1}{2}\right)^2 + \left(x + \frac{1}{2}\right)^3 + \dots$
28. S is the sum of n terms of a geometric progression, P is the product of the n terms and R is the sum of the reciprocals of the terms. Prove that $\left(\frac{S}{R}\right)^n = P^2$.
29. Three unequal numbers a, b, c are such that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in arithmetical progression. Prove that b, a, c are in arithmetical progression.
30. Prove that the G.P $1 + \frac{2x}{3+x^2} + \left(\frac{2x}{3+x^2}\right)^2 + \dots$ is convergent for all values of x and find the limit of its sum.
31. Prove by induction that:
- (i) $3^{3n+2} + 5^{n+1}$ is divisible by 3
- (ii) $n^3 + 2n$ is divisible by 3.
- (iii) $4^{n+1} + 5^{2n-1}$ is a multiple of 21.
- (iv) $11^{n+2} + 12^{2n+1}$ is divisible by 133.
- (c) $\frac{2}{3}\left(\frac{9}{8}\right)^{11}, \frac{2}{3}\left(\frac{9}{8}\right)^{n-1}$
3. (a) 111.1111
- (b) $\frac{182}{243}$
- (c) $1 - (-2)^n$
- (d) $\frac{a^p(a^{3k} - 1)}{a^3 - 1}$
4. $12\frac{3}{8}, 11\frac{13}{16}, \frac{-9}{16}$
5. (a) 8 (b) 7 (c) 8 (d) $n + 1$ (e) n
6. -5, $2\frac{1}{2}$
7. 3 8. 2, $2\frac{1}{2}$, $167\frac{1}{2}$
9. (a) 9 (b) $\frac{1}{9}$ (c) 10^5
10. $\pm 1\frac{1}{2}$ 11. $\pm 3, \pm \frac{2}{3}$
12. 13 13. 31
14. 6, $13\frac{1}{2}$ 15. 182, $\frac{1}{2}$, -1
16. 24 17. £1,070,000
18. 2 21. 12
22. 7 23. 1023
24. $x < 0$ 25. $\frac{4}{5}, 5$
26. $2\frac{1}{2}, 5, 7\frac{1}{2}, 10$
27. (a) $|x| < 1$ (b) $|x| < 3$
- (c) $|x| > 1$ (d) $\frac{-3}{2} < x < \frac{1}{2}$

Answers

1. (a) 5, 2500, 12500
- (b) $\frac{1}{2}, 3, 1\frac{1}{2}$
- (c) $\frac{1}{3}, 1\frac{2}{3}, \frac{5}{9}$
2. (a) $5 \times 2^{10}, 5 \times 2^{19}$
- (b) $10\left(\frac{5}{2}\right)^6, 10\left(\frac{5}{2}\right)^{18}$